Introduction to Economics

Class 13 — Monopoly; oligopoly and the prisoner's dilemma

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9 November 2026

Plan for today

- Monopoly: the price and quantity a single seller picks
- What a monopoly costs the rest of us
- A price for each buyer; what governments do
- Oligopoly: two sellers and a deal between them
- More sellers; when a deal can hold

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 4's third class. Last Monday ended with a mountain's only cable car, a seller that chooses its own price, and left open which price it picks. Today that question is answered and the cost of the answer to everyone else is measured; after the break two sellers share one market, make a deal and feel the pull to break it, and a real train line that went from one company to three shows what more sellers do to a price.

Slide 2 Plan for today

The class at a glance. Before the break, one seller: the cable car's demand table, why each extra ride adds less to its revenue than its price, the number of rides and the price it picks, and what that choice costs the riders who stay at home. After the break, two sellers: a deal between two farms, why each is tempted to break it and what happens when both do; then why more sellers push a price toward cost, and when a deal can hold anyway.

Two halves with a ten-minute break.

Warm-up: average variable or average total cost?

A workshop sews hoodies: total cost in euros a week

hoodies	0	25	50	75
total cost	400	700	900	1,300

At 0 hoodies it pays only the rent.

1. Average variable cost at 50 hoodies?
2. Average total cost at 50 hoodies?
3. The rent rises by 100 a week. Average variable cost at 50 now?

In pairs, two minutes: three answers; then hands.

Variable cost = total cost − fixed cost (class 11, slides 9–11).

Slide 3 Warm-up: average variable or average total cost?

A warm-up on costs read from a table, with a workshop that sews hoodies. Before class, find the fixed cost in the table, then ask which average leaves it out and which counts it, and what a higher rent can change and what it cannot.

Revealed in class.

Warm-up: which events shift demand to the right?

The demand for doughnuts in a town

1. Muffins, a substitute, become more expensive
2. Doughnuts get cheaper, and more of them are sold
3. Coffee, a complement, gets cheaper
4. The doughnut makers pay less for flour

In pairs, two minutes: four answers; then hands.

Its own price: a move along demand; related goods' prices, incomes, tastes: a shift (class 5, slides 13–14).

Slide 4 Warm-up: which events shift demand to the right?

A warm-up on what shifts a demand curve, with doughnuts in a town. Before class, ask for each event whether buyers want more doughnuts at every price or only the point on an unchanged curve moves, and whose plans the event changes: the buyers' or the sellers'.

Revealed in class.

Madrid–Barcelona by fast train: one company, then three

Revealed in class

Before class: Ouigo (May 2021) and Iryo (November 2022) started running rival trains on this line. Guess what had happened by 2025 to the average fare and to the number of passengers — and why.

Source given in class.

Slide 5 Madrid–Barcelona by fast train: one company, then three

A real market where the number of sellers changed: on the high-speed line between Madrid and Barcelona one company, Renfe, sold every ticket until Ouigo arrived in May 2021 and Iryo in November 2022. Before class, guess what had happened by 2025, compared with 2019, to the average fare paid (total ticket revenue divided by the number of passengers) and to the number of passengers, and ask why two newcomers would matter when they are only two.

Revealed in class.

From the second hundred on, each adds less to revenue than its price: the price falls for all

highest price for all (euros)	13	12	11	**10**	9	**8**	7	6
rides a day	100	200	300	**400**	500	**600**	700	800
revenue a day (euros)	1,300	2,400	3,300	**4,000**	4,500	**4,800**	4,900	4,800
marginal revenue (euros a ride)	13	11	9	**7**	5	**3**	1	−1

Marginal revenue: what one more ride adds to revenue
(here: per ride, for each extra hundred)

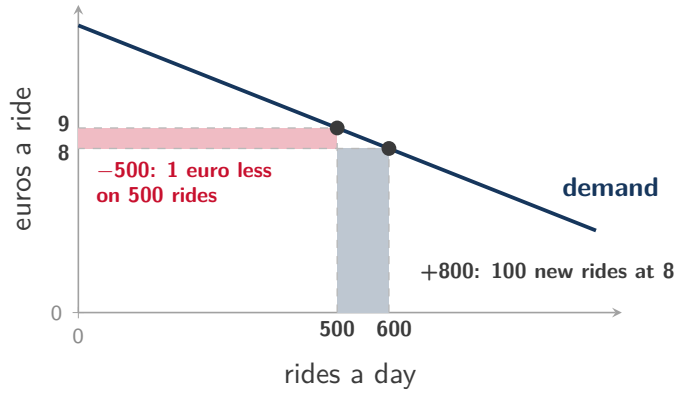
Monopoly: the only seller of a good with no close substitute; it faces the market's demand (class 12, slide 20).

A price taker's extra bag adds just the price: marginal revenue (MR) = P = 4 (class 12, slides 6 and 8).

Slide 6 From the second hundred on, each adds less to revenue than its price: the price falls for all

Back to last Monday's cable car, the only one up its mountain, with the same demand: for each number of rides a day, the highest price at which all of them sell, one price for every rider. Revenue is price times rides; marginal revenue is what one more ride adds to revenue, here measured per ride for each extra hundred. The gap between the top row (price) and the bottom row (marginal revenue) is the whole difference between today and last Monday's print shop: the shop could sell any number of bags at 4 euros, so one more bag added exactly the price; the cable car can sell more only by lowering the price for everyone, so each extra hundred adds less than its price, and past 700 rides it adds less than nothing. The next slide shows where the gap comes from.

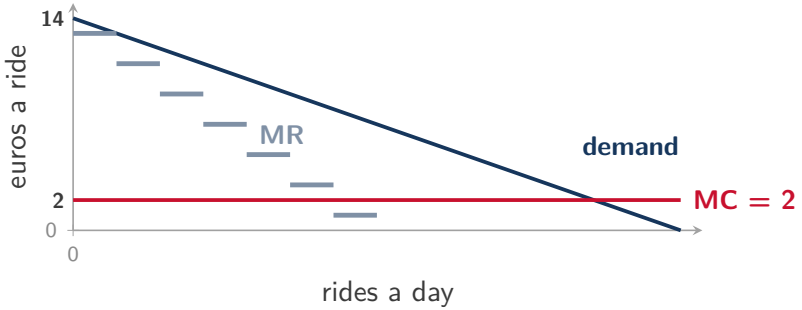
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



Marginal revenue: 800 − 500 = 300 for 100 rides — 3 a ride, below the price of 8.

Which rides does the cable car sell?

Each extra ride costs 2 euros (energy, wear); fixed costs are paid whatever it sells.



Before class: which is the last hundred whose marginal revenue beats the 2-euro cost? What price goes with it?

Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

Slide 7 From 500 to 600 rides: 800 more from new riders, 500 less from the old ones

The gap in slide 6's table, drawn. Moving down the demand curve from 500 rides at 9 to 600 at 8 has two effects that pull against each other: the hundred new riders pay 8 each, but the five hundred who were already riding at 9 now pay a euro less each. Marginal revenue is what is left after the second effect, which is why it sits below the price. A price taker never feels the second effect, because it never has to lower its price to sell more; a monopoly always does. The slide assumes one price for all riders; what the cable car could do if it could charge the old riders 9 and the new ones 8 comes after the break.

Slide 8 Which rides does the cable car sell?

The monopoly's choice. New today is the cost side: each extra ride costs 2 euros, while the cables and cabins cost the same whatever it sells, so they cannot change the best number of rides. The steps are slide 6's marginal revenue, one flat step for each hundred rides. Before class, run along the steps and ask which is the last hundred that adds more to revenue than to cost, and then which price goes with that number of rides on the demand curve.

Revealed in class.

What does the cable car's choice cost?

Revealed in class

Before class: at the cable car's price, some riders who value a ride at more than its 2-euro cost do not ride. How many rides would be sold at a price of 2? Who loses what? (The demand line: price $= 14 -$ rides $\div 100$.)

Slide 9 What does the cable car's choice cost?

The cost of the cable car's choice to riders and cable car together. At its price, some riders who value a ride above its 2-euro cost do not ride. Before class, ask how many rides would be sold at a price of 2 (the demand line: price = 14 minus rides divided by 100), who keeps what at the cable car's price, and what happens to the value of the rides that are not sold.

Revealed in class.

In pairs, four minutes: the only fibre network in town

One firm owns the town's only fibre network; each extra household costs it 10 euros a month.

highest price for all (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

1. Marginal revenue of each extra hundred, per household?
2. Its price, and how many households?
3. Its profit before the fixed cost?
4. How many households should get fibre, town and firm counted together?

In pairs, four minutes: four answers.

Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

Slide 10 In pairs, four minutes: the only fibre network in town

Exercise 1 is slides 6 to 9 on a table, in the notation Problem set 3 uses: one company owns the town's only fibre network and each extra connection costs it 10 euros a month. Before class, work out the revenue at each number of households and what each extra hundred adds per household; then ask which hundreds are worth connecting, what price goes with that number, what the company earns before its fixed cost, and how many connections would be best for the town and company together.

Revealed in class.

The cable car again: which curve moves? (optional)

The cable car (slides 6–8): price = 14 − rides ÷ 100; marginal revenue of each extra hundred: 13, 11, 9, 7, 5, 3, 1 a ride.

1. It raises its price from 10 to 12. Which curve moves?
2. A new road to the summit opens. What happens to its demand?
3. Each ride now costs 4, not 2. Which curve moves?

In pairs, two minutes: three answers.

A price change moves along the demand curve (class 5, slide 10).

Slide 11 The cable car again: which curve moves? (optional)

An optional exercise on what moves which curve, back at the cable car. Before class, ask for each event whether it changes the demand line itself, the cost line, or only the point on an unchanged line.

Revealed in class.

Break

Ten-minute break

Slide 12 Break

Ten minutes. After the break: a different price for each rider, what governments do about a monopoly, then two sellers instead of one.

A different price for each rider?

Revealed in class

Price discrimination: selling the same good at different prices to different buyers.
Before class: if every rider paid exactly what the ride is worth to them, how many rides would the cable car sell, and who would gain the surplus?

Slide 13 A different price for each rider?

Price discrimination means selling the same good at different prices to different buyers. Before class, suppose every rider paid exactly what the ride is worth to them: ask how many rides the cable car would then sell, compared with one price for all, and who would end up with the surplus.

Revealed in class.

Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

- **Competition law:** stop mergers, fine price-fixing
- **Regulation:** cap the price a natural monopoly may charge
- **Public ownership:** the state runs it
- **Doing nothing:** when every cure costs more

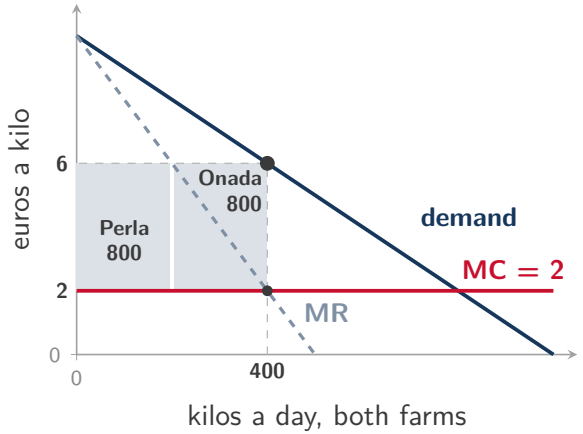
Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

In Spain: the CNMC (Comisión Nacional de los Mercados y la Competencia). Reading Ch. 11, pp. 258–261.

Slide 14 Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

Four things a government can do about a monopoly, each with its own cost. Competition law blocks mergers that would create one and fines firms that fix prices; in Spain the body in charge is the CNMC, the same authority that wrote the train report of slide 5. Regulation caps the price a natural monopoly may charge, such as class 12's electricity grid; the cap cannot simply be set at marginal cost, because a natural monopoly's average cost lies above its marginal cost (the fixed cost is spread over the units), so at that price it would lose money; regulators allow a price near average cost, or a cap that falls year by year to keep the pressure on costs. Public ownership removes the profit motive, and with it much of the pressure to keep costs down. Doing nothing is right when the loss is small and every cure would cost more. Any news story about a regulator or a competition authority fits one of these four boxes.

Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros



Perla and Onada: the only two farms; each kilo costs 2 to farm.

Oligopoly: a market with a few sellers of similar goods

Cartel: sellers who agree to act as one monopoly: here 200 kilos each

Slide 15 Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros

From one seller to two. Perla and Onada are the only farms in the bay, so together they face the market's demand, and each kilo costs either of them 2 euros to farm. A market with a few sellers is an oligopoly, and its mark is that each seller's profit depends on what the other does. If the two acted as one firm they would use the cable car's rule: marginal revenue equals marginal cost at 400 kilos, the price read off demand is 6, and the 1,600 a day of profit splits into 800 each when each farms 200 kilos. An agreement to do this is collusion, and the group is a cartel. It pays them because together they are a monopoly, with a price far above cost: two sellers do not have to compete the price down to cost if they can agree. The next slide asks whether the agreement holds.

Keep the deal or sell more? Each farm's profit a day

Each farm sells 200 kilos (the deal) or 250.
Price: 400 kilos → 6.00 · 450 → 5.50 · 500 → 5.00.

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

profit = (price − 2) × own kilos

Slide 16 Keep the deal or sell more? Each farm's profit a day

The cartel in a payoff table. Each farm picks 200 kilos (the deal) or 250; the price depends on the total, and each cell shows both farms' profit a day, Perla's first, as (price minus 2) times own kilos. Before class, check one cell against the setup line, then take Perla's point of view: for each thing Onada might do, compare Perla's two numbers and ask what Perla would do.

Revealed in class.

Where do the two farms end up?

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

Dominant strategy: best whatever the other does

Nash equilibrium: each does its best, given what the other does

Prisoner's dilemma: each does its best, and both are worse off than if both had cooperated

Slide 17 Where do the two farms end up?

The same table with three terms to learn: dominant strategy, Nash equilibrium, prisoner's dilemma. Before class, for each choice the other farm might make, circle each farm's best reply; look for a cell where both numbers are circled, and compare it with the deal.

Revealed in class.

In pairs, three minutes: open on Sundays?

A town's only two supermarkets: open on Sundays or not? Profit, thousand euros a month:

		Rodó	
		closed	open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

1. Fresc's best if Rodó stays closed?
2. And if Rodó opens?
3. Rodó's best choice?
4. The Nash equilibrium? Who gains?

In pairs, three minutes: four answers; then hands.

Slide 18 In pairs, three minutes: open on Sundays?

Exercise 2 is slide 17 on a new table: two supermarkets, each open on Sundays or closed, with the profits of both in each cell. Before class, take each shop in turn and, for each thing the other might do, compare that shop's own two numbers; then look for the cell neither shop would leave on its own and ask who gains and who loses there.

Revealed in class.

One seller, two, many: where does the price go?

Revealed in class

Before class: as one monopoly the two farms sell 400 kilos at 6 euros; each kilo costs 2. Where does the price go when each farm does its best for itself, and when many farms share the bay? Why?

Slide 19 One seller, two, many: where does the price go?

Back to slide 5's question, with the two farms. As one monopoly they sell 400 kilos at 6 euros, and each kilo costs 2. Before class, ask where the price goes when each farm does its best for itself, and where it goes when many farms share the bay; then ask why a seller with a small share of the market is keener to sell more than a monopoly is.

Revealed in class.

Meeting every day: can the deal hold?

Revealed in class

Before class: Perla breaks the deal on day 1, and from day 2 both sell 250 kilos every day. How much does Perla gain on day 1, and lose on each day after? Tit-for-tat: start with the deal, then do what the other did yesterday.

Slide 20 Meeting every day: can the deal hold?

The two farms do not play once but meet every morning at the market. Before class, suppose Perla breaks the deal on day 1 and from day 2 both sell 250 kilos every day: work out from slide 16's table how much Perla gains on day 1 and loses on each day after, and how many days it takes for the gain to be used up. Tit-for-tat means: start with the deal, then do what the other did yesterday.

Revealed in class.

In pairs, three minutes: a stall at the town fair?

At the town fair, a toy chain, Mega, and a small toy shop, Petita, each choose a stall or not. Profit that day, thousand euros:

		Petita: stall	Petita: no stall
Mega	stall	Mega 9, Petita 2	Mega 10, Petita 3
	no stall	Mega 7, Petita 5	Mega 8, Petita 4

1. Does Mega have a dominant strategy?
2. Petita's best reply to each of Mega's?
3. The Nash equilibrium?

In pairs, three minutes: three answers.

Slide 21 In pairs, three minutes: a stall at the town fair?

Exercise 3 is a game where the two players are not in the same position: a big toy chain and a small toy shop, each deciding on a stall at the town fair. Before class, check for each player whether one choice is best whatever the other does, and if only one of them has such a choice, ask what the other should expect and do about it.

Revealed in class.

Cartels and train fares: positive or normative? (optional)

1. *"Cartels raise the prices buyers pay."*
2. *"Firms in a cartel should pay bigger fines."*
3. A and B both want cheap train tickets. A: *"more companies on every line."* B: *"one public company with capped fares."* Values or a prediction?

In pairs, two minutes: three answers.

Positive: a claim evidence can test · normative: a claim about what should be (class 4, slide 7).

Slide 22 Cartels and train fares: positive or normative? (optional)

An optional exercise on positive and normative claims, with today's topics. Before class, ask of each claim whether evidence could in principle test it or whether it rests on a judgement of what should be, and for the third item whether the two people differ in their goals or in what they expect to happen.

Revealed in class.

A season pass: what does one more trip cost? (optional)

Berta bought a cable-car season pass in October: 150 euros, no refunds. A trip takes an hour she could spend on a lifeguard shift paid 15 euros.

1. Her opportunity cost of one more trip today?
2. Should she ride more "to get her money's worth"?
3. Next autumn: is a new pass's 150 part of her choice?

In pairs, two minutes: three answers.

Opportunity cost: what you give up; sunk: paid and gone whatever you choose (class 2, slide 23).

Slide 23 A season pass: what does one more trip cost? (optional)

An optional exercise on opportunity cost and sunk costs, with a cable-car season pass bought in October. Before class, ask what Berta gives up by taking one more trip today, whether money already paid and not refundable can be part of that, and whether the same holds for a pass she has not bought yet.

Revealed in class.

A monopoly sells while marginal revenue beats marginal cost and reads its price off the demand curve.

Its price is above marginal cost: rides worth more than their cost go unsold.

Slide 24 In one line

The class in four lines, to read slowly: the first is slides 6 to 8, the second slide 9. The third line (slides 15 to 17) and the fourth, which answers slide 5's question, are revealed in class.

Before Thursday 12 November

- Thursday 12 November: monopolistic competition; problems on Blocks 3–4 — read Mankiw & Taylor Ch. 12, pp. 267–276
- Problem set 3: bring your worked solutions on Thursday 12 November (not handed in)
- Practise in Sextant: the units the course page lists



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Slide 25 Before Thursday 12 November

Thursday 12 November: monopolistic competition, many sellers each with its own slightly different product, and problems on Blocks 3 and 4. Reading: Mankiw & Taylor chapter 12, pages 267 to 276. Problem set 3, on Block 4, is not handed in: work it on your own paper and bring your worked solutions, with your copy of the set, to class on Thursday 12 November. Its problems 4 and 5 use today's two moves, the demand table and the two-by-two game. Practise in Sextant with the units the course page lists.

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